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SARDAR PATEL UNIVERSITY

B.Sc. (CA & IT) (SEMESTER-III) EXAMINATION-2019

November 26, 2019, Tuesday

2:00 p.m. to 4:00 p.m.

PS03SIIT22 (Operations Research I)

Maximum Marks: 35

Q.1 Multiple choice question.

[05]

- (1) The main origin of Operations research was during the
 (a) First world War (b) Second world War (c) Third world War (d) None of these
- (2) minimize $z =$
 (a) maximize (z) (b) maximize ($-z$) (c) -maximize ($-z$) (d) none of these
- (3) The coefficient of artificial variable in problem objective function of maximization problem is
 (a) $+M$ (b) $-M$ (c) 0 (d) None of these
- (4) The Transportation Problem is said to be unbalanced if
 (a) $\sum a_i \neq \sum b_i$ (b) $\sum a_i = \sum b_i$ (c) $z_j - c_j \geq 0$ (d) $z_j - c_j \leq 0$
- (5) In MODI method every loop has number of cells.
 (a) odd (b) even (c) prime (d) none of these

Q.2 Answer the following in short. (Attempt any 5)

[10]

- (1) Define Operation research.
- (2) Write the standard form of Linear Programming problem,
 Maximize $z = 3x_1 + 2x_2$
 Subject to constraints $2x_1 + 23x_2 \leq 4$; $15x_1 + 21x_2 \leq 5$; $x_1, x_2 \geq 0$.
- (3) Define slack variables.
- (4) Write dual of the following L.P.P.
 Maximize $z = x_1 + 25x_2$
 Subject to constraints $x_1 + 4x_2 \leq 10$; $2x_1 + 3x_2 \leq 17$; $x_1, x_2 \geq 0$.
- (5) What is transportation problem?
- (6) Write mathematical form of transportation problem.
- (7) Define loop in MODI method.

Q.3 Solve the following L.L.P. using graphical method,

[05]

Maximize $z = 3x_1 + 5x_2$
 Subject to constraints $x_1 + 2x_2 \leq 20$; $x_1 + x_2 \leq 15$; $x_2 \leq 8$; $x_1, x_2 \geq 0$.

OR

Q.3 Solve the following L.L.P. using simplex method,

[05]

Maximize $z = 3x_1 + 5x_2$
 Subject to constraints $x_1 + x_2 \leq 4$; $3x_1 + 2x_2 \leq 18$; $x_1, x_2 \geq 0$.

Q.4 Solve the following L.L.P. using Big M method,

[05]

Maximize $z = 3x_1 - x_2$
 Subject to constraints $2x_1 + x_2 \geq 2$; $x_1 + 3x_2 \leq 3$; $x_1, x_2 \geq 0$.

OR

Q.4 Write dual of the following L.P.P.

[05]

Maximize $z = 3x_1 + 5x_2$
 Subject to $x_1 + x_2 \geq 1$; $x_1 + x_2 \leq 15$; $x_2 \leq 8$; $x_1, x_2 \geq 0$.

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(P.T.O.)

Q.5 Find the initial basic feasible solution for the T.P. using North west corner method.

[05]

	A	B	C	D	Supply
1	6	3	5	4	22
2	5	9	2	7	15
3	5	7	8	6	8
Demand	7	12	17	9	

OR

Q.5 Find the initial basic feasible solution for the T.P. using Least cost method.

[05]

	D_1	D_2	D_3	D_4	Supply
S_1	6	4	1	5	14
S_2	8	9	2	7	16
S_3	4	3	6	2	5
Demand	6	10	15	4	

Q.6 Find the optimal solution of following Transportation Problem using MODI method.

[05]

	Warehouse				
Factory	W_1	W_2	W_3	W_4	Supply
F_1	19	30	50	10	7
F_2	70	30	40	60	9
F_3	40	8	70	20	18
Demand	5	8	7	14	

OR

Q.6 Find initial basic feasible solution of Unbalanced Transportation Problem using Vogel's approximation method.

[05]

	I	II	III	Supply
A	7	10	5	90
B	12	9	4	50
C	7	3	11	80
D	9	5	7	60
Demand	120	100	110	

